

Communication Systems Lab 1

Pranav Chandra
2023AAPS0013P

August 5, 2025

1 Question 1

Question: Generate a rectangular waveform using the fourier series expansion of a rectangular wave at $f_0 = 50Hz$.

$$m(t) = \frac{4}{\pi} \sum_{n=1,3,5,\dots}^N \frac{1}{n} \sin(2\pi n f_0 t) \quad (1)$$

Plot the time domain $m(t)$ and frequency domain $|M(f)|$ for different N .

Answer: This is a relatively simple problem. It was solved in MATLAB. Sampling frequency was set to $2 \times$ the maximum frequency $N \times f_0$. After running the code, figures 1a and 1b were generated for the time domain and frequency domain signals.

The code for this question can be found in Appendix A.

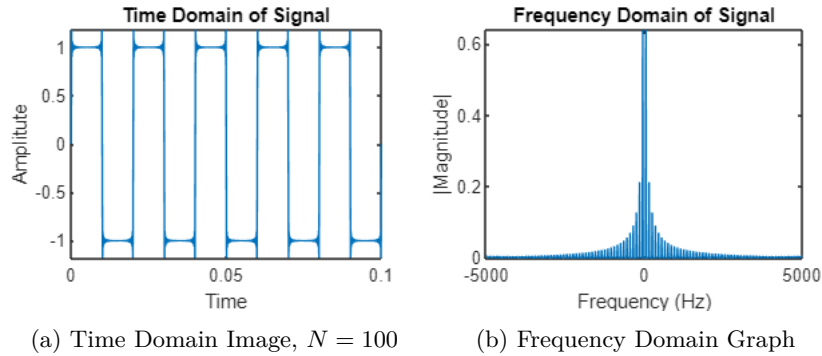


Figure 1: Graphs From Question 1

2 Question 2

Question: Generate a periodic rectangular pulse of period 20ms using the built-in square function and plot its waveform in the time domain and frequency spectrum using fft. Then, pass the rectangular pulse through a low-pass filter with impulse response $2B\text{sinc}(2Bt)$ and observe the output in the frequency domain for different values of B . Use the convolution operation.

This is another line that I'm going to type here.

Answer: Using the square function, Figures 2a and 2b were generated for the time domain and frequency domain representations.

The code for this question can be found in Appendix B.

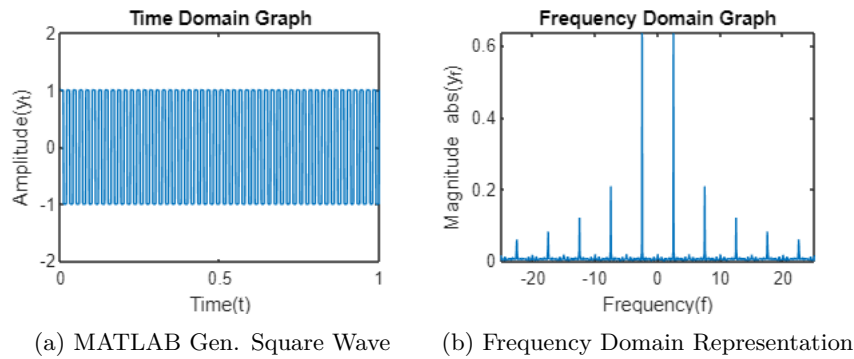


Figure 2: Graphs of the Square Wave

After this, a function `y_convd` was created along with the variable B . Various values of B were iterated upon and the filter was used on the original square wave using the `conv(a, b, 'same')` operation. Figure 3 shows the effect of the convolution. Different values of B and ranges of t showed slightly different effects, but all of them contained the same structure.

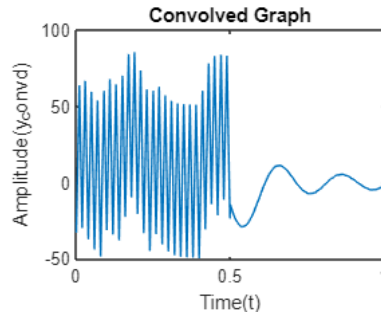


Figure 3: Convolved Function

This concludes the first lab. Further labs will likely be done in Python as opposed to MATLAB.

A Question 1 Code

```
1 N = 100;
2 f = 50;
3 fs = 2*f*N;
4 dt = 1/(fs);
5 t = 0:dt:0.1;
6 m_t = zeros(size(t));
7
8 for n = 1:2:N
9     m_t = m_t + (1/n)*(sin(2*pi*n*f*t));
10 end
11
12 m_t = (4/pi).*m_t;
13
14 m_f = fftshift(fft(m_t));
15 m_f = m_f/norm(m_f);
16 f_range = linspace(-fs/2, fs/2, length(m_f));
17
18 figure();
19 plot(t, m_t);
20 xlabel("Time");
21 ylabel("Amplitude")
22 title("Time Domain of Signal")
23
24 figure();
25 plot(f_range, abs(m_f));
26 xlabel("Frequency (Hz)");
27 ylabel("|Magnitude|");
28 title("Frequency Domain of Signal");
29
```

B Question 2 Code

```
1 %Defining Variables
2 fs = 20*(1/20e-3);
3 t = 0:1/fs:1;
4 B = 5;
5 f = 1/20e-3;
6
7 %Calculating Square Wave
8 y_t = square(2*pi*f*t);
9
10 %Calculating FFT of Wave
11 y_f = fftshift(fft(y_t));
12 y_f = y_f / norm(y_f)
13 N = length(y_t);
14 f_rng = linspace(-f/2, f/2, N);
15
16 %Creating a convolved signal of y and the signal mentioned in the
    qn
17 conv_func = 2*B*sinc(2*B*t);
18 y_convd = conv(y_t, conv_func, 'same');
19
20 %Plotting the various waves
21 figure();
22 plot(t,y_t);
23 xlabel("Time(t)");
24 ylabel("Amplitude(y_t)");
25 ylim([-2 2])
26 title("Time Domain Graph");
27
28 figure();
29 plot(f_rng, abs(y_f));
30 xlabel("Frequency(f)");
31 ylabel("Magnitude abs(y_f)");
32 title("Frequency Domain Graph");
33
34 %Plotting the convolved signal
35 figure();
36 plot(t, y_convd);
37 xlabel("Time(t)");
38 ylabel("Amplitude(y_convd)");
39 title("Convolved Graph");
40
```